Collaborative Filtering

Using other users' behaviour to fill the empty cells

Session 2 | 11:00–11:45 | Theory

ACM Student Chapter Workshop

Where we are and what we need



From Lab 1

- 610 users $\times$ 9 724 movies
- 98.3% of cells empty
- users rate differently (user bias)
- a few blockbusters dominate (popularity)

The question for this session

How can we predict what a user might like using other users' behaviour?

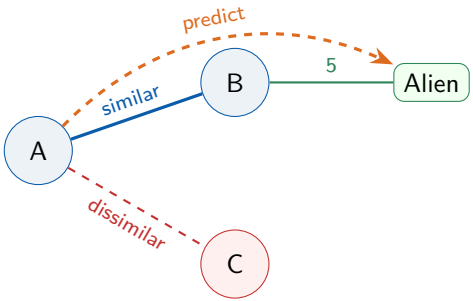
Output of Lab 2

A working recommender that returns Top-N movies for a chosen user.

The core intuition

	LOTR	Matrix	Titanic
A	5	5	1
B	5	4	1
C	1	1	5

- A and B agree on all three films
- C has the opposite taste
- B liked *Alien*, A has not seen it



Prediction

B's opinion is evidence for A: **A will probably like Alien**. C's opinion is evidence *against*.

Two families of collaborative filtering

Memory-based (neighbourhood)

Keep the ratings; at prediction time find neighbours.

- **User-based**: similar users
- **Item-based**: similar items
- no training, simple, explainable

Lab 2 (today, 11:45)

Model-based (latent factor)

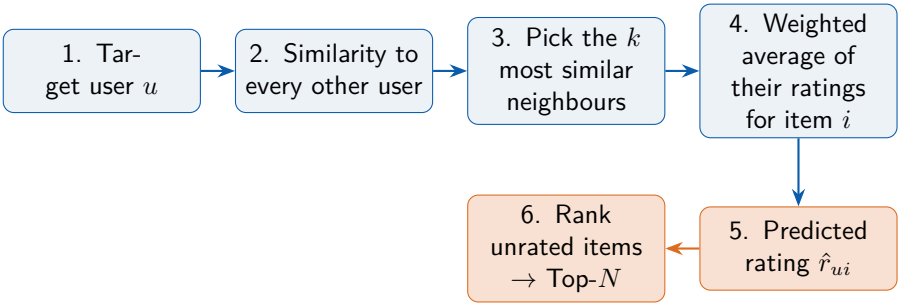
Learn a compact model from the ratings.

- **Matrix factorization** (SVD, ALS)
- clustering, probabilistic models, neural nets
- scales better, handles sparsity better

Lab 3 (this afternoon, 14:15)

We start with the simplest idea, discover its limits, and then fix them.

User-based CF: the pipeline



Lab 2 follows exactly these six boxes. Ingredients we still need: **similarity** (box 2) and the **prediction rule** (boxes 4–5).

Measuring similarity: cosine

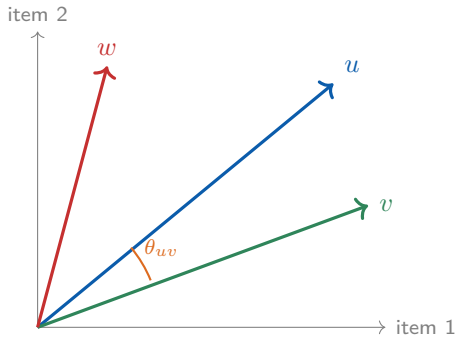
Treat each user as a vector of ratings:

$$\cos(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|} = \frac{\sum_i r_{ui} r_{vi}}{\sqrt{\sum_i r_{ui}^2} \sqrt{\sum_i r_{vi}^2}}$$

- measures the angle, not the length
- $+1$ same direction, 0 orthogonal, -1 opposite
- only items rated by both contribute to the numerator

By hand ($A = (5, 5, 1)$, $B = (5, 4, 1)$):

$$\cos = \frac{25 + 20 + 1}{\sqrt{51}\sqrt{42}} = \frac{46}{46.28} \approx 0.994$$



$\cos \theta_{uv}$ large: similar
$\cos \theta_{uw}$ small: different

A trap: raw cosine on ratings, and the fix

Raw ratings

	LOTR	Matrix	Titanic	cos with A
A	5	5	1	1.00
B	5	4	1	0.99
C	1	1	5	0.40

C is *opposite* to A yet cosine says 0.40: all ratings are positive, so all vectors point in a similar direction.

Mean-centred $(r - \bar{r}_u)$

	LOTR	Matrix	Titanic	cos with A
A	$+1.33$	$+1.33$	-2.67	1.00
B	$+1.67$	$+0.67$	-2.33	0.97
C	-1.33	-1.33	$+2.67$	-1.00

Mean-centring

Subtract each user's average rating, then take the cosine.

above my average $\rightarrow$ positive
below my average $\rightarrow$ negative

Why it matters

It removes the user bias you saw in Lab 1 (generous vs harsh raters) so that we compare *taste*, not *habit*.

Similarity measures at a glance

Measure	Formula	Use
Cosine (raw)	$\frac{\sum_i r_{ui} r_{vi}}{\|r_u\| \|r_v\|}$	implicit / binary data
Pearson correlation	$\frac{\sum_{i \in I_{uv}} (r_{ui} - \bar{r}_u)(r_{vi} - \bar{r}_v)}{\sqrt{\sum_{I_{uv}} (r_{ui} - \bar{r}_u)^2} \sqrt{\sum_{I_{uv}} (r_{vi} - \bar{r}_v)^2}}$	explicit ratings, user-based
Adjusted cosine (items)	$\frac{\sum_u (r_{ui} - \bar{r}_u)(r_{uj} - \bar{r}_u)}{\sqrt{\sum_u (r_{ui} - \bar{r}_u)^2} \sqrt{\sum_u (r_{uj} - \bar{r}_u)^2}}$	explicit ratings, item-based
Jaccard	$\frac{\|I_u \cap I_v\|}{\|I_u \cup I_v\|}$	who-interacted-with-what

I_{uv} = items rated by both u and v. **In Lab 2:** mean-centred cosine (user) and adjusted cosine (item).

Worked example: who is Alice's neighbour?

	LOTR	Matrix	Titanic	Notebook	$\bar{r}$	Pearson
Alice	5	4	1	?	3.33	–
Bob	5	5	2	1	3.25	**+0.97**
Carol	4	5	1	2	3.00	**+0.89**
Dave	1	2	5	5	3.25	−1.00
Eve	2	1	4	4	2.75	−0.84

Task: predict Alice's rating of *The Notebook*.

- similarities use the 3 films Alice has rated (co-rated items)
- Bob and Carol agree with Alice; Dave and Eve are opposites

Raw cosine for comparison

Bob 0.99, Carol 0.98, Dave 0.51, Eve 0.61

Raw cosine cannot tell that Dave is Alice's *opposite*.

Caveat

Only 3 co-rated films: fragile evidence. Real data is worse.

From neighbours to a prediction

$$\hat{r}_{ui} \;=\; \bar{r}_u \;+\; \frac{\sum_{v \in N_k(u,i)} \mathrm{sim}(u,v)\,(r_{vi} - \bar{r}_v)}{\sum_{v \in N_k(u,i)} |\mathrm{sim}(u,v)|}$$

- $N_k(u,i)$: the k most similar users to u *who rated item* i
- $r_{vi} - \bar{r}_v$: how much v liked i relative to v's own average
- weights: more similar $\Rightarrow$ more influence
- add back $\bar{r}_u$: translate to u's own scale

Simplest version (no centring)

$$\hat{r}_{ui} = \frac{\sum_v \mathrm{sim}(u,v)\, r_{vi}}{\sum_v |\,\mathrm{sim}(u,v)|}$$

Key idea: similar users' opinions contribute more.

A weighted *average* keeps the prediction on the rating scale (clip to $[0.5, 5]$).

Worked prediction: Alice and *The Notebook*

Neighbour v	$\mathrm{sim}(A, v)$	$r_{v,\text{Notebook}}$	$\bar{r}_v$	deviation	$\mathrm{sim} \times$ deviation
Bob	0.971	1	3.25	-2.25	-2.184
Carol	0.885	2	3.00	-1.00	-0.885
Sum	$\sum \lvert \mathrm{sim} \rvert = 1.855$				-3.069

$$\hat{r}_{A,\text{Notebook}} = \underbrace{3.33}_{\bar{r}_A} + \frac{-3.069}{1.855} = 3.33 - 1.65 = \mathbf{1.68}$$

- Alice's own average is 3.33, but Bob and Carol found the film *well below* their averages
- So we predict a rating well below hers: **don't recommend Notebook to Alice**

Compare

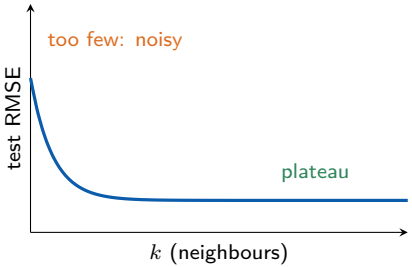
Plain average of Bob and Carol: 1.5
Using all four users ($k = 4$, Dave and Eve get negative weights): 1.75
Item-based CF (later): 1.78

Choosing the neighbourhood

- **Top-k**: the k most similar users who rated the item
- **Threshold**: only $\mathrm{sim}(u, v) > \tau$ (we use > 0)
- **Minimum overlap**: ignore pairs with fewer than m common items

The role of k (bias–variance)

k small: noisy, trusts a few people (high variance)
k large: averages many weakly similar users (high bias, drifts toward popular)



Schematic. Lab 2 §5 draws the real curve: error falls until $k \approx 10$–20, then flattens.

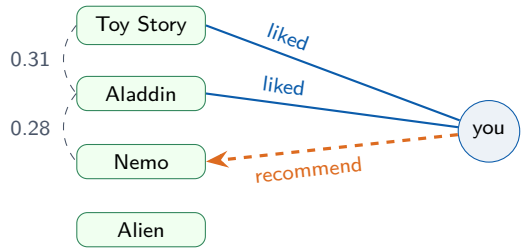
Item-based CF: flip the question

User-based asks: "Which users are similar to *me*?"

Item-based asks: **"Which items are similar to the items I already liked?"**

1. compute item–item similarity from the columns of the matrix
2. for target item i, take the k items most similar to i *that user* u *has rated*
3. weighted average of u's own ratings on them

Similar because the same people rated them alike: no genre data needed.



Nemo is similar to the films you liked, so it is recommended.

Item-based prediction (same worked example)

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_{j \in N_k(i,u)} \mathrm{sim}(i,j)\,(r_{uj} - \bar{r}_u)}{\sum_{j \in N_k(i,u)} |\,\mathrm{sim}(i,j)|}$$

Item j rated by Alice	adj. cosine with Notebook	Alice's deviation $r_{Aj} - 3.33$	product
LOTR	-0.964	$+1.67$	-1.607
Matrix	-0.923	$+0.67$	-0.615
Titanic	$+0.904$	-2.33	-2.109
Sum	2.791 (abs.)		-4.331

$$\hat{r}_{A,\text{Notebook}} = 3.33 + \frac{-4.331}{2.791} = 3.33 - 1.55 = \mathbf{1.78}$$

Notebook is *anti-correlated* with the action films Alice loved and correlated with Titanic, which she hated. Item-based and user-based agree: do not recommend.

User-based vs item-based

	User-based	**Item-based**
Similarity between	users (rows)	items (columns)
Prediction uses	ratings of similar *users* on i	the target user's ratings on similar *items*
Typical m vs n	many users $\Rightarrow$ large $m \times m$ matrix	$n \times n$ matrix, items change slowly
Stability	changes whenever a user rates	more stable, precompute offline
Explanation	"people like you liked it"	"because you liked X"
Serendipity	higher (surprising matches)	lower (safe, similar items)
Cold start	new user: fails	new item: fails

In practice item-based CF often performs better and scales better when there are far more users than items (the Amazon setting).

Limitations of neighbourhood CF ⇒ why matrix factorization?

Problem	What goes wrong	In the lab
Sparsity	few co-rated items ⇒ similarity from 2–3 movies	neighbour table
Cold start	no history ⇒ no neighbours	new-user demo
Scalability	$O(m^2)$ or $O(n^2)$ similarities; searching neighbours at query time	loops over users
Popularity bias	popular items have most data and dominate lists	Top-5 lists
Unreliable similarity	neighbour rated an item ⇒ prediction 5.0 from *one* voice	"without guards"
No explicit bias	user/item effects are only handled by centring	user-mean baseline

Motivating question for the next session

Can we compress users and items into a few **hidden factors** so that every cell has a prediction?

Practical fixes you can apply today

Significance weighting

Shrink similarity when the overlap is small:

$$\mathrm{sim}'(u,v) = \mathrm{sim}(u,v) \cdot \min\Big(1, \tfrac{|I_{uv}|}{\gamma}\Big)$$

Damping / support threshold

$$\hat{r}_{ui} = \bar{r}_u + \frac{\sum_v \mathrm{sim}\,(r_{vi} - \bar{r}_v)}{\lambda + \sum_v |\,\mathrm{sim}\,|}$$

or require $\geq s$ neighbours and $\geq p$ ratings for the item.

Neighbourhood hygiene

positive similarities only; cap k; minimum overlap

Baseline-adjusted CF

Use $b_{ui} = \mu + b_u + b_i$ instead of $\bar{r}_u$ and model the *residual* (this idea leads straight to matrix factorization with biases).

Always validate

Not every published trick helps on every dataset: test significance weighting yourself in Lab 2 (E1).

Prepare for Lab 2: what you will build (11:45–12:45)

1. **Similarity on a toy example**: raw vs mean-centred cosine
2. **User-based CF**: neighbours of user 10 $\rightarrow$ prediction $\rightarrow$ Top-5 titles
3. **Experiment**: $k = 2,\ 20,\ 50$ neighbours
4. **Item-based CF**: movies similar to *Toy Story*
5. **Evaluate**: RMSE against baselines
6. **Cold start**: pretend to be a new user

Look out for

- similarity resting on 2–3 movies
- 5.0 predictions from one neighbour
- $k = 20$ vs $k = 50$: why identical?
- item lists vs user lists

Bring

`train.csv`, `test.csv` from Lab 1
Notebook: `Lab2_Collaborative_Filtering.ipynb`

Summary and check questions

- CF predicts a user's missing ratings from the **behaviour of similar users or items**
- **User-based**: similar users; **item-based**: similar items
- Similarity: mean-centred cosine / Pearson (adjusted cosine for items)
- Prediction: weighted average of neighbours' deviations + own mean
- Limits: sparsity, cold start, scalability, popularity bias, unreliable similarity

Check yourself

1. What is collaborative filtering?
2. User-based vs item-based?
3. How is similarity used to make a recommendation?
4. Why mean-centre before computing cosine?

References and further reading I

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